

Tutorial 2. Camera Models, Filters, Events and Edges

Solutions

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Camera models

Learning Objectives:

- Understand fundamental principles of camera optics and lens systems
- Analyze event-based camera operation and constant brightness assumption
- Master composite optical system modeling using ray tracing matrices
- Apply geometric optics principles to real-world camera configurations

Key Concepts: Thin lens formula, field of view, geometric optics, ray tracing, event-based cameras.

1.

Thin lens formula 1: A camera is equipped with a lens with focal length $f = 50$ mm and aperture diameter $A = 2$ cm. An object at 5 m from the camera is in focus. What is the minimum thickness of the camera? Is the aperture diameter needed for this calculation?

Solution. The thickness of the camera is lower bounded by the distance image plane to the lens plane. According to the thin lens law, we know that

$$\frac{1}{5} + \frac{1}{D'} = \frac{1}{50 \cdot 10^{-3}} \quad (1)$$

where D' is the distance between the image plane and the lens plane. Then, we have $D' = 5.05$ cm. The aperture diameter A is not needed: it affects light collection and defocus blur, but not the in-focus image distance given by the thin-lens equation.

2.

Thin lens formula 2: A camera is equipped with a lens with focal length $f = 40$ mm and aperture diameter $A = 2$ cm. The distance between the image plane and the lens plane is 5 cm. What is the blur diameter of an object at infinity?

Similar triangles for defocus blur: $(A/2)/f = (B/2)/(D' - f)$

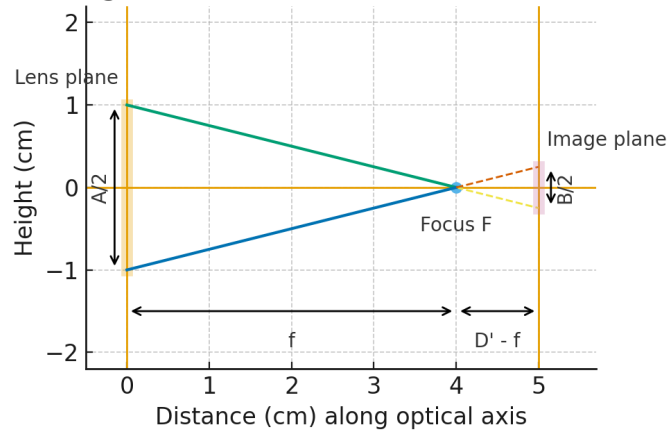


Figure 1. Similar triangles relationship for blur diameter calculation.

Solution. By similar triangles (see Fig. 1), we have that the blur diameter B satisfies

$$\frac{A/2}{f} = \frac{B/2}{D' - f}, \quad (2)$$

where D' is the distance between the image plane and the lens plane.

Thus,

$$B = \frac{A}{f}(D' - f) = \frac{2 \cdot 10^{-2}}{40 \cdot 10^{-3}}(5 \cdot 10^{-2} - 40 \cdot 10^{-3}) = 0.5 \text{ cm} \quad (3)$$

3. **Field of view:** A simple camera is made of a CCD sensor and a single lens. The size of a CCD sensor is $4 \text{ cm} \times 4 \text{ cm}$. The focal length of the lens is $f = 50 \text{ mm}$. What is the field of view (FOV) of this camera?

FOV Geometry: $2\phi = 2 \arctan(d/(2f)) \approx 43.60^\circ$

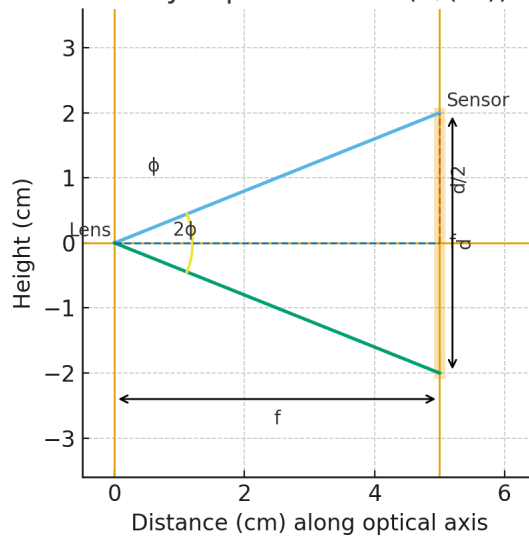


Figure 2. Field of view geometry for a camera with CCD sensor and lens.

Solution. The FOV can be computed according to (see Fig. 2)

$$2\phi = 2 \arctan \frac{d}{2f} = 2 \arctan \left(\frac{4 \cdot 10^{-2}}{2 \cdot 50 \cdot 10^{-3}} \right) \simeq 43.6^\circ, \quad (4)$$

where d is the size of the sensor.

4.

Geometric optics: A simpler way to handle lenses and to determine the imaging equations is to use a *ray representation*. Instead of considering the 3D point in space, we consider a ray in space (a line) and how this is deflected by a lens when it falls within its aperture.

Let us consider a lens with focal length f and a line incident at an angle θ with the lens (the angle is with respect to the normal to the lens plane) at a (signed) height d from the optical center. Compute the exiting angle ϕ by using the two basic rules of a thin lens:

- (a) rays through the optical center are not deflected;
- (b) objects at infinity are focused at a distance f behind the lens (this means that parallel rays converge at a point at a distance f behind the lens)

See Fig. 3.

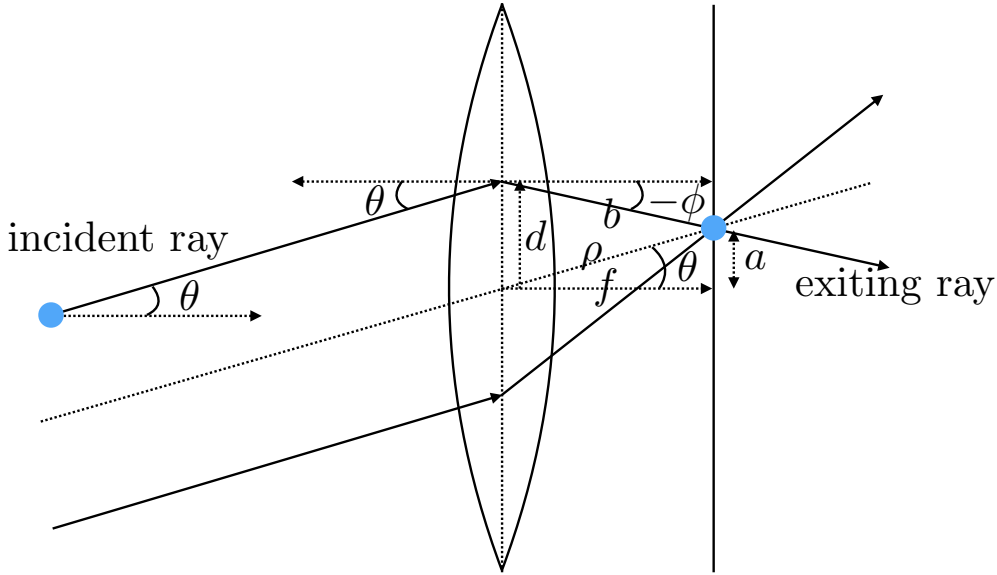


Figure 3. Geometric optics.

Solution. If we take the ray with the same incident angle θ through the optical center, we know that it will not be deflected behind the lens.

Also, the deflected ray from the incident one at a (signed) height d will intersect the previous one at a distance f . In equations, we have

$$f = \rho \cos \theta \quad (5)$$

$$a = \rho \sin \theta = f \tan \theta \quad (6)$$

where ρ is the length of the segment behind the lens from the optical center to the point in focus and a is the (signed) height (on the lens plane) of the point in focus from the optical

center.

$$d - a = -b \sin \phi \quad (7)$$

$$f = b \cos \phi = \frac{a - d}{\tan \phi} \quad (8)$$

where b is the length of the segment of the deflected ray behind the lens up to the point in focus. So now we obtain

$$\tan \phi = \frac{f \tan \theta - d}{f} = \tan \theta - \frac{d}{f} \quad (9)$$

The equation above is the update to the direction of a ray after going through the aperture of a lens.

Let us test the equation with a few cases.

- If $\theta = 0$ the rays are all parallel to the optical axis. In this case $\tan \phi = -\frac{d}{f}$
- Another example is when $d = 0$, *i.e.*, when the ray goes through the optical center and it is not deflected. Then $\theta = \phi$ which is correct in our notation.
- For a converging lens ($f > 0$): when d increases, ϕ decreases; when θ increases, ϕ increases (within the paraxial range $[-\pi/2, \pi/2]$).

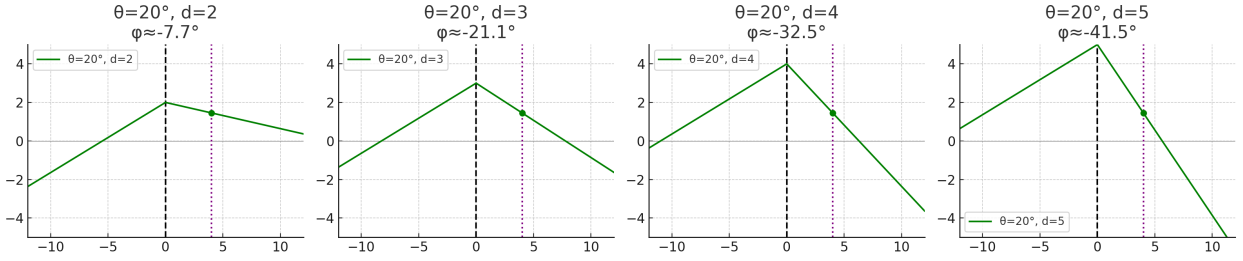


Figure 4. Effect of height variation: as d increases, ϕ decreases for fixed θ .

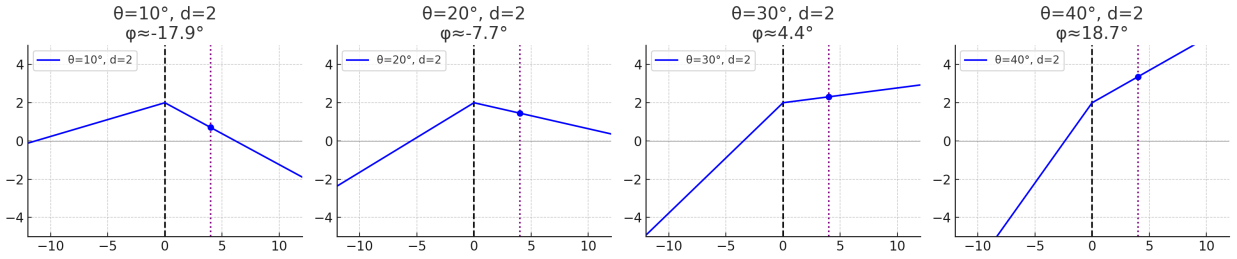


Figure 5. Effect of angle variation: as θ increases, ϕ increases for fixed d .

To make our geometric optics calculations complete, let us represent a ray with the 3D vector $[\tan \theta \quad d \quad 1]^\top$ where θ is the incident angle and d the height (with respect to the optical axis). Then, we already know how to update the 3D representation when it meets a lens with focal length f . We can use the matrix

$$M_L = \begin{bmatrix} 1 & -\frac{1}{f} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (10)$$

So that we have

$$V_{ext} = M_L V_{inc} = \begin{bmatrix} 1 & -\frac{1}{f} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tan \theta \\ d \\ 1 \end{bmatrix} = \begin{bmatrix} \tan \theta - \frac{d}{f} \\ d \\ 1 \end{bmatrix} \quad (11)$$

If the lens is not centered at the optical axis, but its center has height b from the optical axis, we can easily generalize the matrix M_L to

$$M_L = \begin{bmatrix} 1 & -\frac{1}{f} & \frac{b}{f} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (12)$$

as what matters in the lens calculation is the height from the lens center ($d - b$).

We also consider empty space as an optical element. If we have an empty space of Δ , we can update the ray with this matrix

$$M_S = \begin{bmatrix} 1 & 0 & 0 \\ \Delta & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (13)$$

so that we obtain

$$V_{ext} = M_S V_{inc} = \begin{bmatrix} 1 & 0 & 0 \\ \Delta & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tan \theta \\ d \\ 1 \end{bmatrix} = \begin{bmatrix} \tan \theta \\ d + \Delta \tan \theta \\ 1 \end{bmatrix} \quad (14)$$

A ray moving in space does not deflect, but its height changes.

With these few building blocks we can perform approximate *ray tracing*, that is we can simulate the path of a ray as it goes through a combination of lenses and space.

- Composite optical systems (challenging):** Let us consider a camera with 2 lenses and let us use the geometric optics approximation to determine the imaging equations.
5. Suppose that the first lens is centered at the optical axis and has focal length f_1 . The second lens is behind the first lens at a distance Δ_1 , has focal length f_2 and its center is displaced at a height b from the optical axis.

For simplicity we assume that both lenses have infinite aperture. The camera sensor is behind the second lens at a distance Δ_2 . A point P is in front of the lens and emits light rays (see Fig. 6).

Use the optical axis as the z -axis, place the first lens at $z = 0$, and let the object coordinate satisfy $P_Z < 0$. Thus, the propagation distance from P to the first lens is $-P_Z$. The distances Δ_1 and Δ_2 are the physical free-space separations drawn in the figure. Determine the ray tracing matrix M that propagates a ray state from P to the sensor. If the ray state is instead initialized at the first-lens plane, omit the initial propagation from P .

Solution. Let

$$S(\ell) = \begin{bmatrix} 1 & 0 & 0 \\ \ell & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad L(f, b) = \begin{bmatrix} 1 & -1/f & b/f \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (15)$$

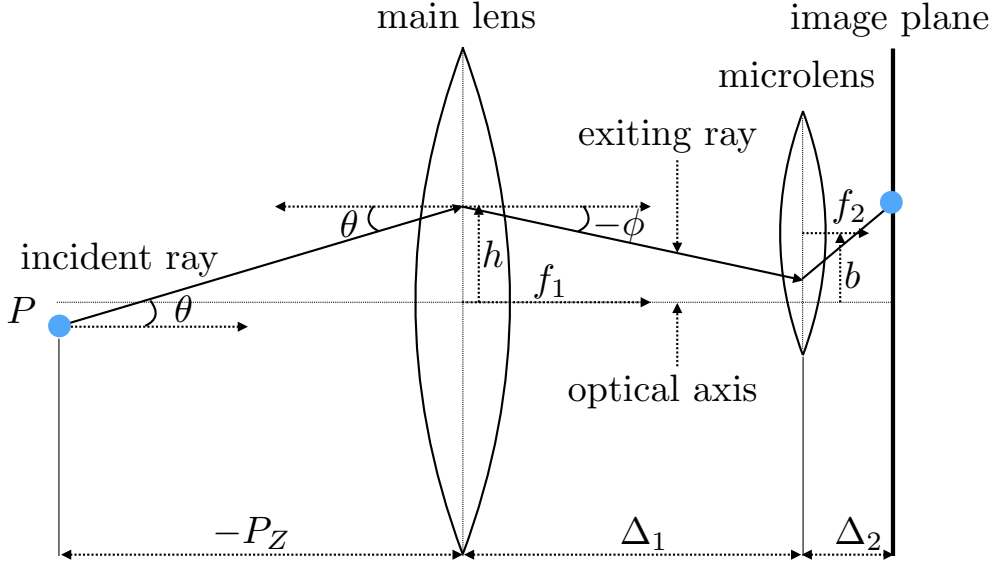


Figure 6. Geometric optics with multiple lenses.

The matrices act from right to left. Starting at the object point P , the ray first propagates by the physical distance $-P_Z$ to the first lens, then passes through lens 1, propagates by Δ_1 , passes through the displaced lens 2, and finally propagates by Δ_2 to the sensor. Therefore,

$$M = S(\Delta_2)L(f_2, b)S(\Delta_1)L(f_1, 0)S(-P_Z). \quad (16)$$

In particular, the free-space factors are $S(\Delta_1)$ and $S(\Delta_2)$ because Δ_1 and Δ_2 are already the physical separations defined in the question; no focal length is subtracted from either distance.

For a compact explicit form, define

$$A = 1 - \frac{\Delta_1}{f_2}, \quad B = \frac{\Delta_1 - f_1 - f_2}{f_1 f_2}, \quad (17)$$

$$C = \Delta_1 + \Delta_2 A, \quad D = 1 - \frac{\Delta_1}{f_1} + \Delta_2 B. \quad (18)$$

The matrix from the first-lens plane to the sensor is

$$M_{\text{system}} = \begin{bmatrix} A & B & b/f_2 \\ C & D & \Delta_2 b/f_2 \\ 0 & 0 & 1 \end{bmatrix}. \quad (19)$$

Consequently, the full matrix from P is

$$M = M_{\text{system}}S(-P_Z) = \begin{bmatrix} A - P_Z B & B & b/f_2 \\ C - P_Z D & D & \Delta_2 b/f_2 \\ 0 & 0 & 1 \end{bmatrix}. \quad (20)$$

If the input ray state is defined at the first-lens plane rather than at P , the appropriate matrix is simply M_{system} .

6.

Event-based camera: Using the constant-brightness assumption and the contrast-threshold event model, derive the event-triggering condition and explain why events are primarily caused by moving edges. State when a strong edge may nevertheless fail to trigger an event.

Solution. An event is generated when the following condition is satisfied:

$$|\log I(x, y, t + \Delta t) - \log I(x, y, t)| = |L(x, y, t + \Delta t) - L(x, y, t)| = |\Delta L| \geq C \quad (21)$$

Derivation from constant brightness assumption:

$$\text{Constant brightness: } L(x(t), y(t), t) = \text{const} \quad (22)$$

$$\Rightarrow \frac{dL}{dt} = \frac{\partial L}{\partial x} \frac{dx}{dt} + \frac{\partial L}{\partial y} \frac{dy}{dt} + \frac{\partial L}{\partial t} = \underbrace{\begin{bmatrix} L_x & L_y \end{bmatrix}}_{\nabla_{\mathbf{x}} L} \underbrace{\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix}}_{\dot{\mathbf{x}}} + L_t = 0. \quad (23)$$

$$\text{(Chain rule: total derivative of } L(x, y, t) \text{ with respect to time)} \quad (24)$$

$$\Rightarrow L_t = -\langle \nabla_{\mathbf{x}} L, \dot{\mathbf{x}} \rangle \quad \text{(Rearranging to express temporal derivative).} \quad (25)$$

First-order Taylor approximation and substitution of Eq. (23) For a fixed pixel (x, y) , the brightness change can be approximated as:

$$\begin{aligned} \Delta L &= L(x, y, t + \Delta t) - L(x, y, t) \simeq L_t \Delta t \stackrel{(23)}{=} -\langle \nabla_{\mathbf{x}} L, \dot{\mathbf{x}} \rangle \Delta t \\ &= -\langle \nabla_{\mathbf{x}} L, \underbrace{\dot{\mathbf{x}} \Delta t}_{\Delta \mathbf{x}} \rangle = -\langle \nabla_{\mathbf{x}} L, \mathbf{v} \Delta t \rangle. \end{aligned} \quad (26)$$

where:

- $\dot{\mathbf{x}} = (u, v)^\top$ is the image plane velocity (optical flow), $\mathbf{v} \equiv \dot{\mathbf{x}}$
- $\Delta \mathbf{x} = \mathbf{v} \Delta t$ is the displacement in the pixel plane over time Δt
- The dot product captures only the motion component along the gradient direction

Conclusion: Moving edges trigger events From Eq. (26), we have:

$$|\Delta L| \approx |\nabla_{\mathbf{x}} L \cdot \mathbf{v}| \Delta t \geq C.$$

This leads to the following behavior:

- $\nabla_{\mathbf{x}} L = \mathbf{0}$ (uniform regions) $\Rightarrow$ no events generated
- $\nabla_{\mathbf{x}} L \perp \mathbf{v}$ (gradient orthogonal to motion) $\Rightarrow \Delta L \approx 0 \Rightarrow$ events unlikely
- $\nabla_{\mathbf{x}} L \parallel \mathbf{v}$ (gradient parallel to motion) $\Rightarrow |\Delta L|$ maximum $\Rightarrow$ events most likely to exceed threshold C

Therefore, events are primarily caused by **moving edges**, where the brightness gradient is aligned with the motion direction.

Additional notes

- The triggering condition uses $|\Delta L| \geq C$, where $\text{sign}(\Delta L)$ determines the *event polarity* (brightening/darkening)
- This first-order approximation assumes sufficiently small Δt
- The constant brightness assumption neglects rapid changes in reflection or illumination

Filters

Learning Objectives:

- Master various filtering techniques and their mathematical foundations
- Understand convolution operations and filter design principles
- Analyze advanced filtering methods like non-local means and mean shift
- Apply filtering concepts to noise reduction and image enhancement

Key Concepts: Convolution, median filtering, averaging filters, non-local means, mean shift, kernel design.

1. **Median filter** [Python implementation is available¹]: The median filter takes the middle element after sorting the elements in a patch. While this operation is quite robust to non-Gaussian noise, it can lead to a severe flattening of the texture when the size of the patch is large. How could we modify the median filter so that it retains the original texture while removing outliers?

Solution. One possible modification is a quartile-based heuristic. For each neighborhood, compute its first quartile, median, and third quartile. Keep the center pixel unchanged when it lies inside the interquartile range; replace only a center pixel below the first quartile or above the third quartile by the neighborhood median. This retains more texture than replacing every center pixel by the median. It is one possible modified-median rule, not a unique standard definition.

2. Consider the 1×7 image **I** below, where each pixel intensity is denoted with a letter $a, \dots, g$:

a	b	c	d	e	f	g
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- (a) Apply a 1×3 averaging filter twice to the image I , that is, apply the averaging filter to I to obtain a smoothed image J and then apply the same filter again to J . Use zero padding to calculate the convolution of edge pixels if needed. Give the expression for the center pixel of the resulting image as a function of the original pixel values in I . Show all your calculations.

Solution. The mean filter

$$K = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \quad (27)$$

The values of c , d , and e after the filter has been applied.

$$c_1 = \frac{b + c + d}{3} \quad (28)$$

$$d_1 = \frac{c + d + e}{3} \quad (29)$$

$$e_1 = \frac{d + e + f}{3} \quad (30)$$

The value of d_1 after the filter has been applied to it.

$$d_2 = \frac{c_1 + d_1 + e_1}{3} = \frac{b + 2c + 3d + 2e + f}{9} \quad (31)$$

- (b) Define a single 1D filter that, when applied only once to the image I , will produce the same result as applying the previous 1×3 averaging filter twice. Specify the dimension of the filter and its values.

Solution.

1/9	2/9	3/9	2/9	1/9
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It is the convolution of the 1×3 averaging filter with itself.

3.

Given the image patches I and I' , find a horizontal 1D convolutional filter k that transforms the image patch I into I' . Assume the patch I is padded with zeros (just one value on each side of the patch), so the valid output size of the convolution $I * k$ matches the size of I' . Show your calculations.

$$I * k = I'$$

$$\begin{bmatrix} 128 & 128 & 128 \\ 64 & 192 & 64 \\ 0 & 128 & 64 \end{bmatrix} * k = \begin{bmatrix} 96 & 128 & 96 \\ 80 & 128 & 80 \\ 32 & 80 & 64 \end{bmatrix}$$

Solution. The filter has a size of 1×3 . Notice that this is a convolution, so the kernel should be flipped horizontally. For the 3rd row of I we will have

$$\begin{aligned} k_3 \cdot 0 + k_2 \cdot 0 + k_1 \cdot 128 &= 32 \\ k_3 \cdot 0 + k_2 \cdot 128 + k_1 \cdot 64 &= 80 \\ k_3 \cdot 128 + k_2 \cdot 64 + k_1 \cdot 0 &= 64 \end{aligned} \quad (32)$$

Thus, $k = [0.25 \quad 0.5 \quad 0.25]$.

4. **Non-local means filter (challenging):** The non-local means filter performs averaging of patches that are sufficiently similar. Under the simplifying assumption that the selected patches share the same clean patch and that their noise terms remain independent, suppose that the original image is affected by zero-mean i.i.d. Gaussian noise. What is the distribution of the denoised patch $\hat{P}$?

If we extract a patch P from the noisy image, then we select other patches P_i such that

$$\|P - P_i\|_F < d \quad (33)$$

with a certain scalar $d > 0$. If N patches P_i satisfy the constraint above, then the updated/denoised patch $\hat{P}$ is obtained via averaging

$$\hat{P} = \frac{1}{N} \sum_{i=1}^N P_i \quad (34)$$

Suppose that the original image is affected by zero-mean i.i.d. Gaussian noise. Then, we can write

$$P = P_0 + n \quad P_i = P_0 + n_i, \quad (35)$$

where P_0 is the shared noise-free patch and $n, n_i \sim \mathcal{N}(0, \sigma^2 I_d)$ are independent Gaussian random variables under this simplified model. In a practical non-local means algorithm, selecting patches according to their observed similarity can introduce statistical dependence; this exercise intentionally ignores that effect.

Solution. For $\hat{P}$ we have

$$\hat{P} = \frac{1}{N} \sum_{i=1}^N P_i = \frac{1}{N} \sum_{i=1}^N (P_0 + n_i) = \frac{1}{N} \left(N P_0 + \sum_{i=1}^N n_i \right) = P_0 + \underbrace{\frac{1}{N} \sum_{i=1}^N n_i}_{\text{Gaussian}} \quad (36)$$

The average of Gaussian random variables is also a Gaussian random variable. Its mean is the average of the means, which are all zero in this case.

$$E(\hat{P}) = E \left(P_0 + \frac{1}{N} \sum_{i=1}^N n_i \right) = E(P_0) + E \left(\frac{1}{N} \sum_{i=1}^N n_i \right) = P_0 + \frac{1}{N} \sum_{i=1}^N \underbrace{E(n_i)}_0 = P_0 \quad (37)$$

Because the selected noise terms are independent, the covariance of their average is

$$\text{Cov}(\hat{P}) = \text{Cov} \left(\frac{1}{N} \sum_{i=1}^N n_i \right) = \frac{1}{N^2} \sum_{i=1}^N \underbrace{\text{Cov}(n_i)}_{\sigma^2 I_d} = \frac{\sigma^2}{N} I_d. \quad (38)$$

Thus, under the stated simplifying assumptions,

$$\hat{P} \sim \mathcal{N} \left(P_0, \frac{\sigma^2}{N} I_d \right). \quad (39)$$

The distribution of the averaged patches is concentrated around P_0 , the shared noise-free patch. This conclusion is conditional on treating the selected noise terms as independent; data-dependent patch selection can violate that assumption in practice.

5.

Mean shift filter (challenging): The *mean shift* filter is a procedure to find the maxima of a probability density function (the modes) from its samples. For the Epanechnikov/hard-neighborhood setting used below, find the relation between one mean shift update and the non-local mean filter above.

The first step is to assume that the probability density function can be approximated by its kernel density estimate

$$p(x) = \frac{1}{N} \sum_{i=1}^N K\left(\frac{|x - x_i|^2}{h}\right), \quad (40)$$

where $K(|x|^2)$ is a kernel that integrates to 1 in x and h is a bandwidth parameter. A necessary condition to determine the maxima of a function is that the first order derivatives must be zero. By taking the first order derivatives in x we obtain

$$\nabla p(x) = \frac{1}{N} \sum_{i=1}^N 2 \frac{x - x_i}{h} K'\left(\frac{|x - x_i|^2}{h}\right) = 0 \quad (41)$$

Next we rearrange the equation above to obtain the following iterative update

$$x^{t+1} = \frac{\sum_{i=1}^N x_i K'\left(\frac{|x^t - x_i|^2}{h}\right)}{\sum_{i=1}^N K'\left(\frac{|x^t - x_i|^2}{h}\right)} \quad (42)$$

This update is the *mean shift* algorithm and it can be shown to converge to the modes of the distribution $p(x)$ in a number of cases.

Let us consider the Epanechnikov kernel

$$K(z) = \begin{cases} \frac{3}{4}(1 - z) & 0 \leq z \leq 1 \\ 0 & z > 1 \end{cases} \quad (43)$$

then, we have

$$K'(z) = \begin{cases} -\frac{3}{4} & 0 \leq z \leq 1 \\ 0 & z > 1 \end{cases} \quad (44)$$

and the meanshift iteration becomes

$$x^{t+1} = \frac{\sum_{i=1}^N x_i \delta(|x^t - x_i|^2 < h)}{\sum_{i=1}^N \delta(|x^t - x_i|^2 < h)} \quad (45)$$

where $\delta(\text{event})$ is the indicator function of «event» (it is 1 if the event is true and 0 otherwise).

Solution. For this Epanechnikov/hard-neighborhood special case, one mean shift update is an unweighted average of the samples inside the current neighborhood. It therefore has the same averaging form as the simplified non-local mean update above. Repeating this update moves the estimate toward a mode. This equivalence does not hold unchanged for arbitrary kernels or weighting rules.

Edges

Learning Objectives:

- Understand the mathematical foundations of gradient-based edge detection
- Analyze self-similarity properties and their relationship to image gradients
- Connect geometric properties to directional derivatives and orthogonality conditions
- Apply gradient analysis to characterize image structure and features

Key Concepts: Image gradients, self-similarity, directional derivatives, orthogonality, edge characterization.

1.

Gradients 1: Let us consider an image I . Let also $p, q \in \mathbb{R}^2$ be two pixels (represented as two 2D vectors) in I . We define *self-similarity* as the property that rays originating at p should have constant image intensity. How does the gradient $\nabla I[q]$ in the point q relate to the self-similarity in the limit as $\alpha \rightarrow 1$?

We define *self-similarity* as the property that

$$I[q] = I[p + \alpha(q - p)] \quad \forall q : \|q - p\|_2^2 \leq \rho \quad (46)$$

where $\rho > 0$ is the radius of a ball around p and $0 < \alpha < 1$. See Fig. 7.

In other words, rays originating at p should have constant image intensity. Notice that, in particular, the self-similarity property is satisfied at corners and at edges.

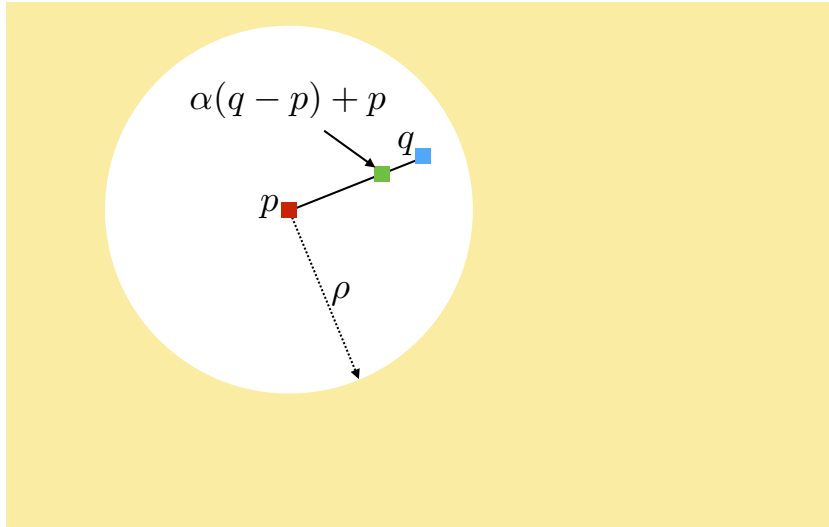


Figure 7. Gradients and self-similarity.

Solution. The self-similarity property is equivalent to

$$I[p + \alpha(q - p)] - I[q] = 0 \quad \forall q : \|q - p\|_2^2 \leq \rho \quad (47)$$

Then, on the one side

$$\lim_{\alpha \rightarrow 1} \frac{I[p + \alpha(q - p)] - I[q]}{1 - \alpha} = 0 \quad (48)$$

The denominator $(1 - \alpha)$ comes from rearranging the self-similarity condition. Since we want to analyze what happens as $\alpha \rightarrow 1$, we divide by $(1 - \alpha)$ to create a difference quotient that reveals the rate of change.

On the other side, this is equal to a [directional derivative](#) with respect to $v = p - q$. Let's work through this step by step:

$$0 = \lim_{\alpha \rightarrow 1} \frac{I[p + \alpha(q - p)] - I[q]}{1 - \alpha} \quad (49)$$

$$= \lim_{h \rightarrow 0} \frac{I[p + (1 - h)(q - p)] - I[q]}{h} \quad (\text{substituting } h = 1 - \alpha) \quad (50)$$

$$= \lim_{h \rightarrow 0} \frac{I[p + q - p - hq + hp] - I[q]}{h} \quad (\text{expanding}) \quad (51)$$

$$= \lim_{h \rightarrow 0} \frac{I[q + h(p - q)] - I[q]}{h} \quad (\text{simplifying}) \quad (52)$$

$$= \lim_{h \rightarrow 0} \frac{I[q + hv] - I[q]}{h} \quad (\text{where } v = p - q) \quad (53)$$

$$= \nabla_v I[q] \quad (\text{definition of directional derivative}) \quad (54)$$

$$= \langle \nabla I[q], v \rangle \quad (\text{directional derivative formula}) \quad (55)$$

$$= \nabla I[q]^\top (p - q) \quad (\text{substituting } v = p - q) \quad (56)$$

$$= -\nabla I[q]^\top (q - p) \quad (\text{factoring out negative sign})$$

Finally, we have $\nabla I[q]^\top (q - p) = 0$ which means the orthogonality of $\nabla I[q]$ and $(q - p)$ or equivalence to zero of one of these vectors.

2. **Gradients 2:** Show that the family of all images I that can be written as a function of only the direction of $v = q - p$, with $v \neq 0$, satisfies the orthogonality property below for a fixed pixel p .

Show that the family of all images I that can be written as a function of only the direction of $v = q - p$

$$I[q] = f\left(\frac{v}{\|v\|_2}\right) = f\left(\frac{q - p}{\|q - p\|_2}\right) \quad (57)$$

for any function f , satisfies the property below for a fixed pixel p

$$\nabla I[q]^\top (q - p) = 0 \quad \forall q \neq p \quad (58)$$

Solution. Let us represent the function f in the form

$$f\left(\frac{v}{\|v\|_2}\right) = f(g(v)), \quad (59)$$

where $g(v) = \frac{v}{\|v\|_2} = \begin{bmatrix} \frac{v_1}{\sqrt{v_1^2 + v_2^2}} \\ \frac{v_2}{\sqrt{v_1^2 + v_2^2}} \end{bmatrix}$ for $v \neq 0$.

Then

$$\nabla_v f(g(v)) = J_g(v)^\top \nabla f \left(\frac{v}{\|v\|_2} \right) \quad (60)$$

$$= \frac{1}{\|v\|_2^3} \begin{bmatrix} v_2^2 & -v_1 v_2 \\ -v_1 v_2 & v_1^2 \end{bmatrix} \nabla f \left(\frac{v}{\|v\|_2} \right), \quad (61)$$

Applying the definition of the property, we get

$$\begin{aligned} \nabla I[q]^\top (q - p) &= \frac{1}{\|v\|_2^3} \nabla f \left(\frac{v}{\|v\|_2} \right)^\top \begin{bmatrix} v_2^2 & -v_1 v_2 \\ -v_1 v_2 & v_1^2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \\ &= \frac{1}{\|v\|_2^3} \nabla f \left(\frac{v}{\|v\|_2} \right)^\top \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 \quad \forall q \neq p. \end{aligned} \quad (62)$$